

Association of Psychotic Experiences With Subsequent Risk of Suicidal Ideation, Suicide Attempts, and Suicide Deaths

A Systematic Review and Meta-analysis of Longitudinal Population Studies

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IMPORTANCE Recent research has highlighted that psychotic experiences are far more prevalent than psychotic disorders and associated with the full range of mental disorders. A particularly strong association between psychotic experiences and suicidal behavior has recently been noted.

OBJECTIVE To provide a quantitative synthesis of the literature examining the longitudinal association between psychotic experiences and subsequent suicidal ideation, suicide attempts, and suicide deaths in the general population.

DATA SOURCES We searched PubMed, Excerpta Medica Database, Cumulative Index to Nursing and Allied Health Literature, and PsycINFO from their inception until September 2017 for longitudinal population studies on psychotic experiences and subsequent suicidal ideation, suicide attempts, and suicide death.

STUDY SELECTION Two authors searched for original articles that reported a prospective assessment of psychotic experiences and suicidal ideation, suicide attempts, or suicide death in general population samples, with at least 1 follow-up point.

DATA EXTRACTION AND SYNTHESIS Two authors conducted independent data extraction. Authors of included studies were contacted for information where necessary. We assessed study quality using the Newcastle-Ottawa Quality Assessment Scale. We calculated pooled odds ratios using a random-effects model. A secondary analysis assessed the mediating role of co-occurring psychopathology.

MAIN OUTCOMES AND MEASURES Psychotic experiences and subsequent suicidal ideation, suicide attempts, and suicide death.

RESULTS Of a total of 2540 studies retrieved, 10 met inclusion criteria. These 10 studies reported on 84 285 participants from 12 different samples and 23 countries. Follow-up periods ranged from 1 month to 27 years. Individuals who reported psychotic experiences had an increase in the odds of future suicidal ideation (5 articles; $n = 56\ 191$; odds ratio [OR], 2.39 [95% CI, 1.62-3.51]), future suicide attempt (8 articles; $n = 66\ 967$; OR, 3.15 [95% CI, 2.23-4.45]), and future suicide death (1 article; $n = 15\ 049$; OR, 4.39 [95% CI, 1.63-11.78]). Risk was increased in excess of that explained by co-occurring psychopathology: suicidal ideation (adjusted OR, 1.59 [95% CI, 1.09-2.32]) and suicide attempt (adjusted OR, 2.68 [95% CI, 1.71-4.21]).

CONCLUSIONS AND RELEVANCE Individuals with psychotic experiences are at increased risk of suicidal ideation, suicide attempts, and suicide death. Psychotic experiences are important clinical markers of risk for future suicidal behavior.

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There has been extensive research on psychotic experiences (PEs) in the general population over the past 2 decades. These are hallucinatory experiences and delusional beliefs that are similar to the classic positive symptoms of schizophrenia but typically associated with at least some degree of intact reality testing. Psychotic experiences are reported by 5% to 8% of the general adult population.¹ While initial research focused on an increased risk for psychotic disorder in individuals who report PEs,² much subsequent research has demonstrated that PEs are associated with high risk for a broad range of mental disorders and poor mental health outcomes in general.³⁻⁹

A striking finding in recent research on PEs has been the strong association with suicidal behavior. In the past 5 years, there have been many studies to investigate the risk of suicidal behavior in individuals who report having PEs.¹⁰⁻¹⁴ To evaluate PEs as a risk marker for future suicidal thoughts and behavior, we conducted a systematic review and meta-analysis of longitudinal studies looking at PEs and risk for suicidal ideation, suicide attempts, and suicide death.

Method

Search Strategy

Following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines,¹⁵ we conducted a systematic review and meta-analysis of published longitudinal studies on PEs, and subsequent suicidal ideation, suicide attempts, and suicide death (Figure 1). The search strategy was developed in consultation with a research librarian. We searched through electronic databases PubMed, Excerpta Medica Database (Embase), Cumulative Index to Nursing and Allied Health Literature (CINAHL), and PsycINFO from their inception until September 2017 for articles with the following search terms: *psychotic experience*, *psychotic like experience*, *psychosis like experience*, *hallucinations*, *delusion**, *psychotic like symptoms*, and *suicid**. We searched using the format “((“psychotic experiences”[Title/Abstract] OR “psychotic-like experiences”[Title/Abstract] OR “psychotic like experiences”[Title/Abstract] OR “psychosis like experiences”[Title/Abstract] OR “hallucinations”[Title/Abstract] OR “Delusion*”[Title/Abstract] OR “psychotic-like symptoms”[Title/Abstract] OR “psychotic like symptoms”[Title/Abstract])) AND ((Suicid*[Title/Abstract] OR suicide[Mesh]).” References within articles were also searched to identify other possible studies. Members of the author team (I.K., J.D., and M.C.) with expertise in this subject area were canvassed to determine if any relevant studies were missing.

Eligibility Criteria

Inclusion criteria were: (1) an original, published article (2) written in English (3) about general population samples and (4) involving prospective assessment of suicidal ideation, suicide attempts, or suicide death, (5) with at least 1 follow-up point. We excluded articles for the following reasons: they (1) were not longitudinal, (2) did not report odds ratios, hazard ratios,

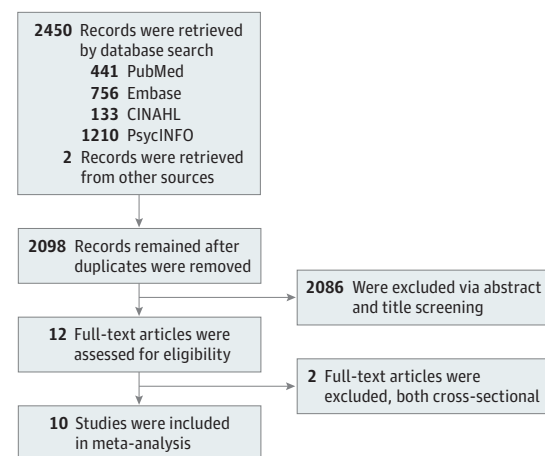
Key Points

Question Are psychotic experiences associated with an increased risk of later suicidal ideation, suicide attempt, and/or suicide death?

Findings In this systematic review and meta-analysis of 10 general population cohort studies on 84 285 individuals, psychotic experiences were associated with significantly increased odds of subsequent suicidal ideation, suicide attempts, and suicide death. The increase in risk was in excess of that explained by co-occurring psychopathology.

Meaning Psychotic experiences are clinical markers of risk for future suicidal behavior in excess of the risk associated with co-occurring psychopathology.

Figure 1. PRISMA Style Flow Diagram of Included Studies



risk ratios, or data that enabled the researchers to calculate these, and (3) were only published as an abstract or conference summary.

Study Selection and Data Extraction

Two authors (K.Y. and U.L.) conducted the literature search. Data extraction was completed separately and any disagreements were resolved via discussion. A senior reviewer (I.K.) was consulted when necessary.

Synthesis of Results

From each included study, these 2 researchers independently extracted data on the baseline age range of participants, the follow-up period, the psychosis and suicidality measures used, and the figures for suicidality outcome (suicidal ideation, suicide attempts, or suicide death; Table). Where necessary, authors were contacted for further information.

To allow for pooled results, we extracted raw unadjusted data from all studies in which data were available and used these to calculate odds ratios for suicidal ideation, suicide attempt, and suicide death in Review Manager (Revman) version 5.3 (Cochrane Collaboration) using a random-effects analysis. Where raw data were not available, we input ORs from the individual studies directly. We also calculated the population

Table. Data Extraction Form of Relevant Data From the 10 Included Studies

Source	Country	Sample, No.	Baseline Age, y	Follow-up Period, y	Instrument	Odds Ratio (95% CI)						
						Unadjusted			Adjusted			
						Psychosis	Suicide	Suicidal Ideation	Suicide Attempt	Suicide Death	Suicidal Ideation	Suicide Attempt
Bromet et al, ²⁹ 2017	19 countries ^a	World Health Organization World Mental Health surveys, 33 370	≥18	NR	Composite International Diagnostic Interview	Composite International Diagnostic Interview	3.90 (3.58-4.26)	3.87 (3.42-4.37)	NA	2.2 (1.8-2.6) ^b	1.9 (1.5-2.5) ^b	NA
Cederlöf et al, ¹² 2017	Sweden	The Child and Adolescent Twin Study in Sweden, 9242	15 or 18	1 mo-5.8 y	Self-report questionnaire	Swedish Patient Register	NA	1.84 (1.23-2.75)	NA	NA	NA	NA
Connell et al, ¹³ 2016	Australia	Mater-University of Queensland Study of Pregnancy, 644	14	16-19	Youth self-report	Structured Clinical Interview for DSM-IV Axis I Disorders	1.60 (1.15-2.24)	1.97 (1.14-3.40)	NA	NA	NA	NA
Fisher et al, ³¹ 2013	New Zealand	Dunedin Multidisciplinary Health and Development study, 789	11	27	Diagnostic Interview Schedule for Children	Diagnostic Interview Schedule	NA	6.45 (2.12-19.62)	NA	NA	2.58 (1.10-6.08) ^c	NA
Honings et al, ²⁵ 2016	Germany	The Early Developmental Stages of Psychopathology study, 3021	21.7	4.8	Composite International Diagnostic Interview	Composite International Diagnostic Interview	2.07 (1.50-2.86)	3.95 (1.85-8.43)	NA	1.32 (0.93-1.87) ^d	3.44 (1.54-7.69) ^d	NA
	The Netherlands	The Netherlands Mental Health Survey and Incidence study, 7076	18-64	3	Composite International Diagnostic Interview	Composite International Diagnostic Interview	NA	NA	NA	NA	NA	NA
	The Netherlands	The Netherlands Mental Health Survey and Incidence study 2, 6646	18-65	3	Composite International Diagnostic Interview	Composite International Diagnostic Interview						
Kelleher et al, ¹⁴ 2013	Ireland	Saving and Empowering Young Lives in Europe, 1112	13-16	1	Adolescent Psychotic Symptoms Screener	Paykel Suicide scale	NA	9.36 (4.40-19.91)	NA	NA	5.90 (3.48-10.01) ^e	NA
Kelleher et al, ²⁸ 2014	Sweden	Swedish Twin Study of Child and Adolescent Development, 2263	13-14	3	Youth self-report	Youth self-report	2.70 (1.40-5.21)	NA	NA	NA	NA	NA
			16-17	3	Adult self-report	Adult self-report	2.72 (1.72-4.29)	NA	NA	NA	NA	NA
Martin et al, ²⁷ 2015	Australia	Helping to Enhance Adolescent Living Project, 1896	12-17	1	Diagnostic Interview Schedule for Children	Self-Harm Behavior Questionnaire, part A	NA	2.48 (1.07-5.75)	NA	NA	NA	NA
Shariffi et al, ³⁰ 2015	United States	Epidemiologic Catchment Area, 15 049	≥18	24-27	Diagnostic Interview Schedule	US National Death index	NA	NA	4.39 (1.63-11.78)	NA	NA	2.28 (0.36-14.41) ^f
Sullivan et al, ²⁶ 2015	United Kingdom	Avon Longitudinal Study of Parents and Children, 3171	12	4	Psychosis-Like Symptom interview	Self-report questionnaire	1.93 (1.50-2.49)	2.37 (1.68-3.37)	NA	1.31 (0.99-1.73) ^g	1.75 (1.20-2.54) ^g	NA

Abbreviations: NA, not applicable; NR, not reported.

^a Argentina, Colombia, Mexico, Peru, Brazil (São Paulo only), United States, Nigeria, Iraq, Lebanon, China (Shenzhen only), New Zealand, Belgium, France, Germany, Italy, the Netherlands, Portugal, Romania, and Spain.

^b Adjusted for age cohorts, sex, person-year dummies, country, and 21 other antecedent mental disorders (mood disorders, anxiety disorder, behavior disorders, eating disorders, and substance use disorders).

^c Adjusted for age, family socioeconomic status, attention deficit/hyperactivity disorder at age 11 years, anxiety, conduct disorder, and depression.

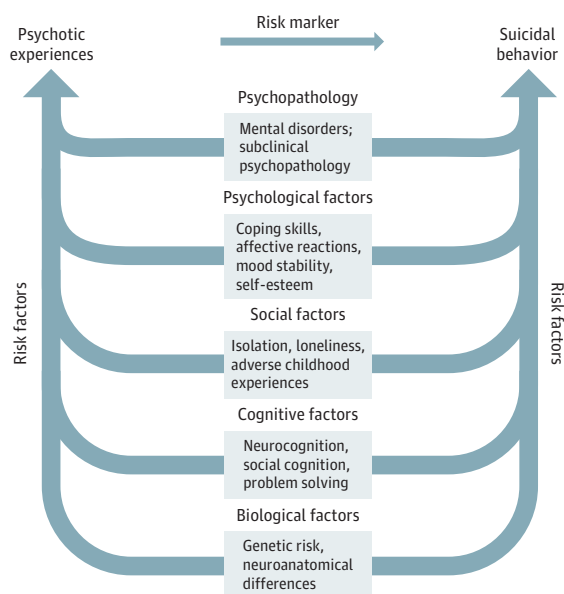
^d Adjusted for age, sex, continuous depression, anxiety, and mania symptoms.

^e Adjusted for abnormal or borderline abnormal score on emotional problems, conduct problems, and hyperactivity subscales of the Strengths and Difficulties Questionnaire.

^f Adjusted for sex, ethnicity, marital status, education, schizophrenia spectrum disorders, depressive disorders, bipolar disorder, phobic disorders, obsessive-compulsive disorder, panic disorder, alcohol use disorders, other substance use disorders, antisocial personality disorder, and any psychiatric hospital admission.

^g Adjusted for sex, social class, and self-reported depressive symptoms at age 12 years.

Figure 2. Conceptual Model of the Association Between a Range of Risk Factors and Psychotic Experiences and Suicidal Behavior



Because of shared risk factors, psychotic experiences function as a risk marker for suicidal behavior.

attributable fraction for the association between PEs and suicidal ideation and between PEs and suicide attempts and suicide deaths combined.

We used τ as a standard deviation and τ^2 to describe the distribution of true effects (ie, as a measure of heterogeneity).^{16,17} The use of τ and τ^2 is considered more appropriate than I^2 for reporting heterogeneity because they provide a point estimate of the between-studies variance, whereas I^2 is the “proportion of variability in the point estimates that is due to τ^2 .”^{18(p1159)} In contrast, τ is the variance of the true effects on the same scale as the effect itself. There is no accepted significance level for τ and τ^2 . Rather, the purpose of τ^2 is to assign weights to studies in a random-effects analysis.

Because the follow-up time differed between included studies, we further conducted a meta-regression to see whether the association between PEs and the risk of suicide ideations or attempts changes over time. For this purpose, we used the mean follow-up time for each study.

Secondary Analysis

We conducted a secondary meta-analysis of studies that had adjusted their findings for co-occurring psychopathology. This was not because we view psychopathology as a confounder in the association between PEs and suicidal behavior; rather, we consider psychopathology an important mediator in the association, and we wished to investigate whether it fully mediates the association (ie, that the reason for PEs being a risk marker for suicidal behavior is solely because PEs are also markers of risk for psychopathology) or just partially mediates the association (ie, that part but not all of the association is explained by co-occurring psychopathology; Figure 2).

Study Quality

Study quality was assessed using the Newcastle-Ottawa Quality Assessment Scale.¹⁹ This assesses studies on 3 categories: selection (including representativeness of the exposed cohort, selection of nonexposed cohort, and ascertainment of the exposure), comparability (on the basis of design or analysis), and outcome (assessments of outcome, whether follow-up was long enough for outcomes to occur and the adequacy of the follow up of cohorts). Exclusion of individuals with baseline suicidal ideation or suicide attempt, the use of semistructured interviews or validated self-report questionnaires, and the reporting of raw, unadjusted data were criteria for assessments of quality as higher. Quality was assessed from the published articles, supplement materials, and information gathered from the authors of individual studies. The comparability category, which offers a maximum score of 2 for adjusting for 2 variables deemed important (by the researchers evaluating a study, to define the adjustments important to their specific research), is not applicable in this systematic review and meta-analysis. Therefore, we modified the scale so that the maximum total quality score was 8 and offered a score of 1 for this category if articles provided unadjusted odds ratios or the raw data necessary to calculate these. In line with previous research using this scale, we defined a high-quality study as any that scored 6 or more (eTable in the Supplement).²⁰⁻²²

Results

Study Selection

Our literature database search yielded 2540 articles, and additional means yielded 2 more. After the removal of duplicates, titles, and (as necessary) abstracts were read to determine articles of relevance. References of included studies were also checked. This identified 12 articles for screening of the full texts. Two of 12 were excluded because they were cross-sectional in design,^{23,24} leaving 10 articles for inclusion in the review (Figure 1). Of these, 7 studies reported odds ratios (ORs),^{13,14,25-29} 2 studies reported hazard ratios,^{12,30} and 1 reported relative risk.³¹

Sample Characteristics

These 10 studies reported on a total of 84 285 participants from 12 different samples and 23 countries (Argentina, Colombia, Mexico, Peru, Brazil [São Paulo only], United States, Nigeria, Iraq, Lebanon, China [Shenzhen only], New Zealand, Belgium, France, Germany, Italy, the Netherlands, Portugal, Romania, Spain, Ireland, Sweden, Australia, and the United Kingdom; Table). In the 6 studies that reported sex information, there were 18 241 women (58.3%).

A total of 5920 individuals had suicidal ideation from the 2 studies that reported the raw numbers for this outcome^{26,28} (of a total 40 111 individuals [14.8%]). Across the 5 studies that provided the raw numbers for suicide attempts,^{12,14,26,29,31} there was a total of 2208 suicide attempts (of 47 684 individuals [4.6%]). One article looked at completed suicides

and reported 24 suicide deaths in their sample of 15 049 individuals.

The follow-up periods from the 10 studies ranged from 1 month to 27 years. The earliest study looked at data from 1980,³⁰ and the latest studies reported baseline data from 2009.^{12,14} Nine of 10 studies were rated high-quality (eTable in the [Supplement](#)).

Meta-analytical Results

Suicidal Ideation

Five studies reported longitudinal data on suicidal ideation. The τ^2 value for studies on suicidal ideation was 0.20. Because there were fewer than 10 studies in this category, it was inappropriate to conduct the Egger test for funnel plot asymmetry³²; however, we have included a funnel plot nonetheless to allow visual inspection (eFigure 1 in the [Supplement](#)). As the 2014 study by Kelleher et al²⁸ reported data for 2 age ranges at 2 different points (with PEs in individuals aged 13 to 14 years associated with suicidal ideation at age 16 to 17 years and PEs in individuals aged 16 to 17 years associated with suicidal ideation at ages 19 to 20 years), both ORs were entered into the model. Individuals who reported PEs had a 2.39-fold (95% CI, 1.62-3.51) increased odds of suicidal ideation.

Suicide Attempts

Eight studies reported longitudinal data on suicide attempts. The τ^2 value for studies on suicide attempts was 0.15. Because there were fewer than 10 studies in this category, it was inappropriate to conduct the Egger test for funnel plot asymmetry³²; however, we constructed a funnel plot to allow visual inspection (eFigure 1 in the [Supplement](#)). Individuals who reported PEs had a 3.15-fold (95% CI, 2.23-4.45) increased odds of suicide attempt.

Suicide Deaths

Just 1 study reported longitudinal data on suicide deaths.³⁰ Therefore we calculated an odds ratio based on the data from this 1 study. Individuals who reported PEs had a 4.39-fold (95% CI, 1.63-11.78) increased odds of suicide death (Figure 2).

Population-Attributable Fractions

The population-attributable fraction of PEs for suicidal ideation was 11.9%. The population-attributable fraction of PEs for suicide attempts and suicide deaths combined was 24.7%. We were not able to estimate CIs because only 2 of the 5 studies on suicidal ideation provided information on the number of individuals with suicidal ideation among those with and without PEs, and 5 of 8 studies did the same for suicide attempts (Figure 3).

Effect of Follow-up Time

Five studies of suicidal ideation and 7 studies of suicide attempts reported a mean follow-up time. Because there were fewer than 10 studies in each category, it was not appropriate to conduct a formal meta-regression for either³³; however, we constructed bubble plots for each category to allow visual inspection (eFigure 2 in the [Supplement](#)).

Psychopathology as a Mediator of the Association Between PEs and Risk of Suicidal Ideation or Suicide Attempt

As a secondary analysis, 3 of 6 studies on the association between PEs and suicidal ideation reported odds ratios adjusted for co-occurring psychopathology; the pooled odds ratio was 1.59 (95% CI, 1.09-2.32). Five of the 8 studies on the association between PEs and suicide attempt reported odds ratios adjusted for co-occurring psychopathology. The pooled odds ratio was 2.68 (95% CI, 1.71-4.21; [Figure 4](#)).

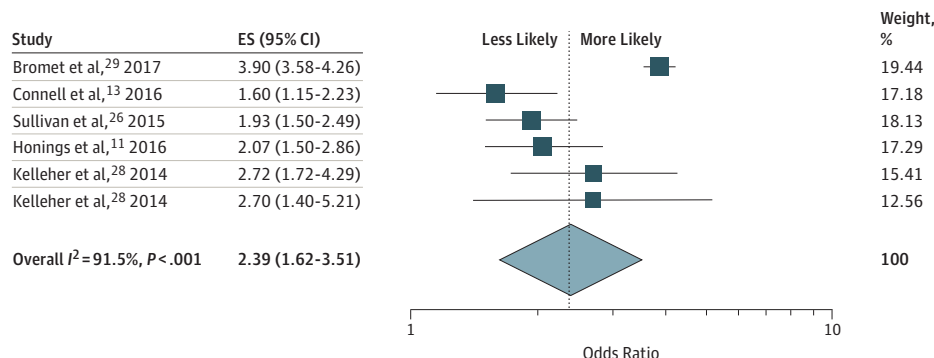
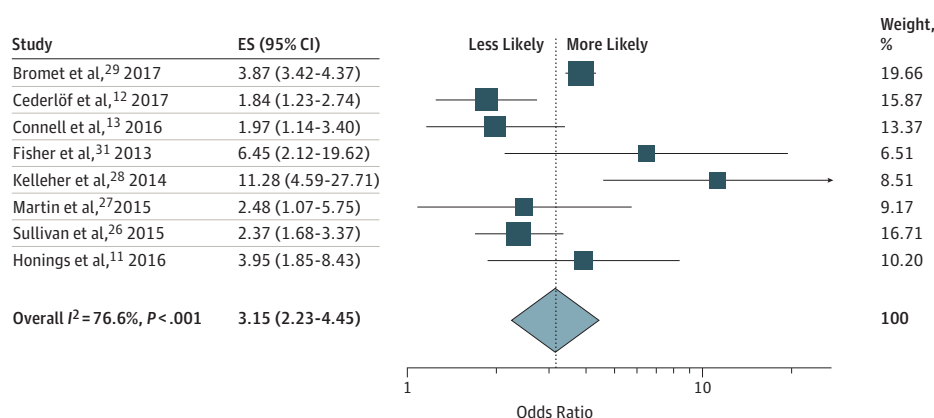
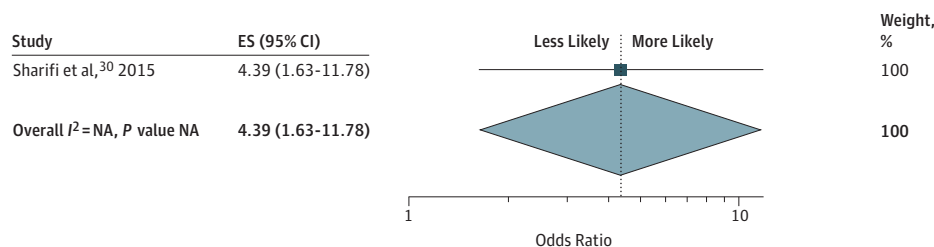
Discussion

An association between psychotic disorder and suicidal behavior has long been recognized; in 1911, Bleuler wrote that “the suicidal drive is the most serious of schizophrenic symptoms.”^{34(p83)} Only recently, however, has research extended this finding to recognize an important association between suicidal behavior and not just psychotic disorders, but also the far more prevalent phenomenon of PEs. We identified 10 prospective cohort studies, all published since 2013, that examined the association between PEs and subsequent suicidal ideation, suicidal attempt, and/or suicide death. All studies were consistent in finding a significant association. Our pooled odds ratios showed that individuals who reported PEs had 2-fold increased odds of subsequent suicidal ideation, 3-fold increased odds of subsequent suicide attempt, and 4-fold increased odds of subsequent suicide death.

The mechanisms explaining the association between PEs and suicidal behavior are potentially manifold.³⁵ One hypothesized direct association between PEs and suicidal behavior is that the content of PEs promotes suicidal behavior (for example, hallucinations commanding the individual to harm themselves). However, the content of hallucinations, including the presence of command hallucinations to harm oneself, cannot be said to emerge at random; rather, hallucination content reflects factors present in the thought content of the affected individual. Therefore, command hallucinations to harm oneself may simply reflect underlying suicidal thoughts (as opposed to inducing de novo suicidal thoughts). Regardless, 2 studies in general population samples found that, despite the strong association between PEs and suicidal behavior, direct commands to harm oneself were present in only a minority of cases.²³ It could also be the case that distress associated with PEs increases the risk of suicidal behavior.³⁶ In a sample of Australian adolescents with PEs, Martin et al²⁷ found that individuals with co-occurring psychological distress had a higher odds of suicide attempt but that individuals without co-occurring psychological distress did not.

There are multiple other factors that may mediate the association between PEs and suicidal behavior.³⁷ Psychotic experiences are associated with a high prevalence of mental illness, which is itself an important risk factor for suicidal behavior. In fact, individuals with PEs not only have a higher overall prevalence of mental disorders, they tend to have more severe and multimorbid psychopathology, a higher burden of symptoms, poorer global functioning, and more mental health

Figure 3. Forest Plot Showing Association Between Psychotic Experiences and Suicidality

A Suicidal ideation**B** Suicide attempt**C** Death by suicide

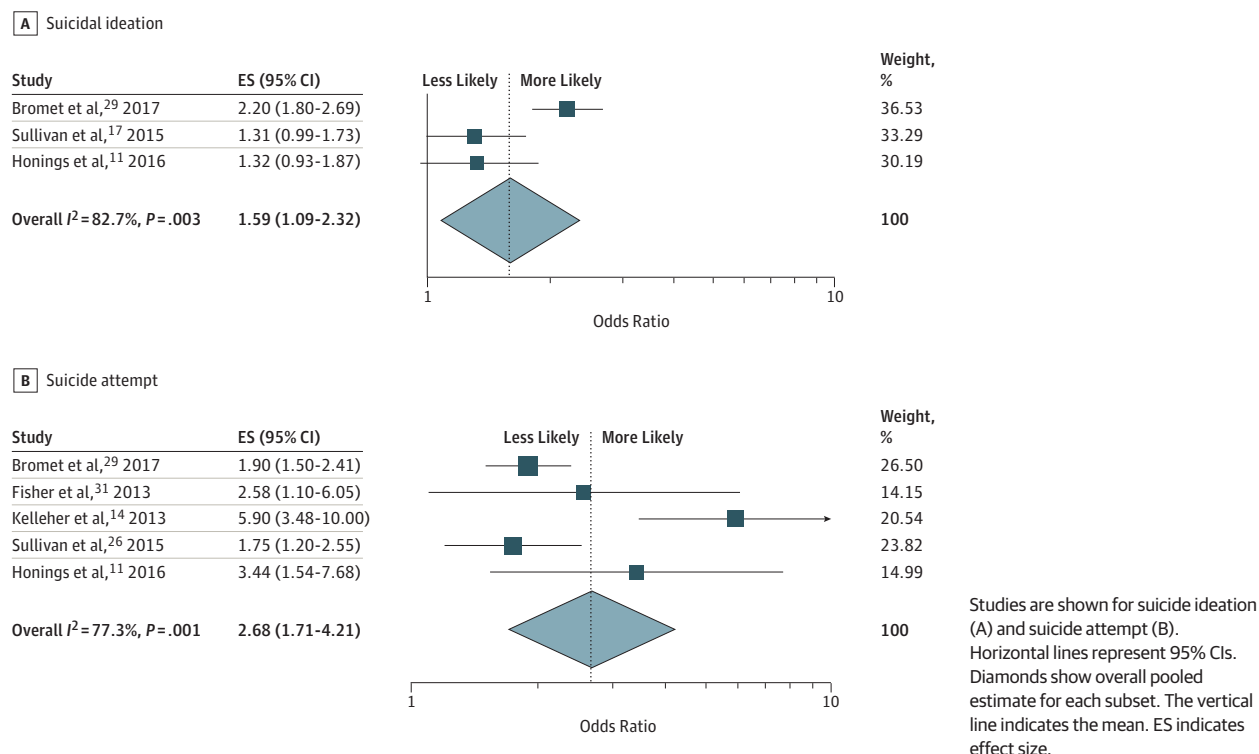
Studies are shown for suicide ideation (A), suicide attempt (B), and suicide death (C). Horizontal lines represent 95% CIs. Diamonds show overall pooled estimate for each subset. The vertical line indicates the mean. ES indicates effect size.

service use than individuals with mental disorders who do not have PEs.^{5,38-40}

However, the presence of mental illness is insufficient to explain the association between PEs and suicidality. For one, PEs are associated with increased odds of suicidal behavior even in individuals without a mental disorder.⁴¹ Furthermore, studies that have adjusted for the presence of co-occurring psychopathology, including both continuous and categorical measures,^{16,17,42,43} have found that risk of suicidal behavior exceeds the level that can be explained by mental illness. In the current study, we conducted a secondary analysis on the association between PEs and suicidal behavior to investigate whether co-occurring psychopathology fully mediated the association. We found that co-occurring psychopathology is only a partial mediator of the association be-

tween PEs and suicidal behavior; that is, PEs are a marker of risk for later suicidal behavior in excess of the risk associated with co-occurring mental disorders. (Figure 2 includes examples of mediating variables in the association between PEs and suicidal ideation and behavior.)

Interestingly, PEs have also been shown to be markers of risk for suicidal behavior in individuals with suicidal ideation. For example, a study by DeVlyder et al¹⁰ showed that in a general adult population sample with suicidal ideation, additional information on depressive symptoms did not help to distinguish who among the individuals with suicidal ideation was at increased odds of suicide attempt, but the co-occurrence of PEs did add significantly to suicide attempt prediction. Similarly, in a general adolescent sample with suicidal ideation, Kelleher et al²³ showed that

Figure 4. Forest Plot Showing the Association Between Psychotic Experiences and Suicidality, Adjusted for Psychopathology

the co-occurrence of PEs was associated with greatly increased odds of suicide attempt.

To our knowledge, there has been only 1 study to date that has looked at PEs and suicidal behavior in a clinical population.²³ This study found that PEs were not associated with a significantly increased odds of isolated suicidal ideation (that is, suicidal ideation in the absence of a suicide plan or suicide attempt), but that PEs were associated with a 3-fold increased odds of suicide attempt. The authors went on to directly compare diagnostic groups with vs without PEs and found that patients with depression and PEs had a 9-fold increased odds of suicide attempt compared with patients with depression who did not have PEs. Patients with an anxiety disorder and PEs had a 15-fold increased odds of suicide attempt compared with patients with an anxiety disorder who did not have PEs. Patients with a behavioral disorder and PEs had a 3-fold increased odds of suicide attempt compared with patients with a behavioral disorder who did not have PEs.

Poorer mathematical and communication skills have also been shown in individuals with PEs,^{44,45} suggesting that problem-solving skills (a well-established risk factor for suicidal behavior⁴⁶) may also be impaired in individuals with PEs. In terms of specific neurocognitive domains, researchers have demonstrated poorer processing speed in particular in individuals with PEs.⁴⁷⁻⁴⁹ This might suggest that additional (time) pressures during mental challenges may elicit cognitive problems that would not be evident under nonpressured settings. This idea is complemented by neuroimaging findings, which have shown reduced integrity of major white matter tracts in

individuals with PEs, which are important in coordinating speeded transmission of information between distributed neural networks.⁵⁰⁻⁵² The real-world consequences of this dysfunction may be that stressful or high-pressure situations precipitate problems with thinking and reasoning that would not otherwise be evident in people who are vulnerable to PEs. In the context of high stress and poorer communication skills, this might adversely affect the individual's ability to formulate logical plans to manage perceived challenges and instead increase the likelihood of turning to suicide.

In a mega-genome wide association study, Pain et al⁵³ demonstrated significant genetic overlap between PEs and both schizophrenia and major depressive disorder risk genes, 2 disorders associated with major risk for suicidal ideation and behavior. Using a convergent functional genomics approach, Niculescu et al⁵⁴⁻⁵⁶ showed that many of the top biomarkers for suicide are also biomarkers for psychotic disorders and high hallucination states. In a series of studies of individuals who were hospitalized for suicidal behavior and postmortem brain tissue from individuals who died by suicide, Niculescu et al found that changes in molecular markers that were associated with high hallucination states, such as *DOCK5*, were also associated with high suicidal states and suggested that a psychosis-like state may be a core feature of suicidality. Other risk factors that may partially mediate the PE-suicidal behavior association include poorer coping skills,⁵⁷ stronger affective reactions to stress,⁵⁸ more mood instability,⁴³ more isolation and loneliness^{59,60} lower self-esteem,³⁷ and higher rates of adverse childhood experiences, such as physical abuse,

sexual abuse, and bullying in individuals with PEs^{24,31,37,61,62} (Figure 2).

Strengths

A notable strength of the review is that we used longitudinal studies, which established a temporal association between PEs and suicidal behavior. All 10 studies were consistent in demonstrating an increased risk. What is more, the findings were consistent in demonstrating increased (albeit varying) levels of risk across a wide variety of countries, including in Africa, Asia, Europe, the Middle East, North America, and South America, suggesting that the association is not culture-specific.

Limitations

Previous research has suggested that suicidal ideation in the absence of suicide attempts (ie, isolated suicidal ideation) may not be associated with PEs—that is, it may be the case that only more severe forms of suicidal behavior, such as attempted and completed suicide, are associated with PEs.^{10,14} We were unable to look at isolated suicidal ideation in the current study, because some individuals who reported suicidal ideation may also have had a suicide attempt and we were unable to identify these individuals. This may have resulted in an overestimate of the effect size of the association between PEs and suicidal ideation outside of the context of more serious suicidal behavior. While participants in the included studies ranged in age from 11 to 65 years, most of the studies involved

adolescents and young adults; we were also unable to look at specific age groups within the studies. Future research should investigate the association between PEs and suicidal behavior in later adulthood specifically. In addition, only 1 study looked specifically at suicide death; further research on PEs and suicide deaths will be valuable. Finally, this review only included articles written in the English language.

Conclusions

Psychotic experiences are a clinical marker of risk for future suicidal behavior, predicting a 2-fold increased odds of subsequent suicidal ideation (5 articles), a 3-fold increased odds of suicide attempt (8 articles) and a 4-fold increased odds of suicide death (1 article). Assessment of PEs (and not just symptoms of a fully psychotic individual) should form an important part of any mental state examination. Our findings suggest that there is a psychosis-associated subtype of suicidal behavior that extends well beyond the previously established association between psychotic disorder and suicidal behavior. Further research is necessary to understand whether specific types of PEs (for example, perceptual abnormalities vs unusual thought content) are more closely associated with suicidal behavior, whether suicidal behavior is a risk factor for later PEs (in addition to PEs being a risk marker for later suicidal behavior), and the interplay between the many potential mechanisms that contribute to the PE-suicidality association.

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